

HIGH ACRYLONITRILE NBR: PATENT LANDSCAPE FOR COMPOSITION-CONTROLLED NITRILE RUBBER AND RELATED LATEX/COMPOSITE PROCESSING

1. ABSTRACT

This report surveys five primary patents that intersect with high-acrylonitrile nitrile butadiene rubber (NBR) synthesis, processing, or use in latex/composite systems. The target is defined as NBR enriched in acrylonitrile, typically pursued to increase oil resistance, chemical resistance, and performance in demanding elastomer applications, while managing tradeoffs in gel content, particle size, processability, and interfacial adhesion. The patent set is heterogeneous: one patent is directly centered on nitrile rubber suspension/latex handling after hydrogenation and precipitation (EP2325238B1), two patents disclose acrylonitrile-containing copolymer latexes used in glove or dip-molding applications (US9388302B2 and US20200299532A1), one patent discloses ABS graft polymerization with explicit acrylonitrile feed control but is only indirectly relevant to NBR chemistry (US8299175B2), and one patent focuses on CNT/graphene/polymer composites that includes a nitrile resin base and a butadiene-acrylonitrile copolymer as a matrix component (EP2561011B1). Across the landscape, the dominant polymerization route remains emulsion polymerization, often with staged monomer addition, emulsifier control, and chain-transfer regulation to tune particle size, gel content, and final latex stability. For high-acrylonitrile NBR specifically, the most relevant evidence emphasizes monomer composition control rather than a dedicated high-ACN NBR recipe: acrylonitrile contents of 30-35 wt% appear in glove latex examples, while the ABS patent uses 4-20 parts by weight acrylonitrile with 17-40 parts styrene and a staged butadiene latex preparation. Process trends include two-step monomer feed strategies, use of emulsifiers and mercaptan chain-transfer agents, coagulation/aging conditions for powder recovery, and latex blending or lamination to balance antistatic behavior, softness, and adhesion. The evidence suggests that practical high-ACN NBR development is most strongly supported by emulsion-polymerized latex systems with controlled feed ratios, moderate temperatures, and downstream coagulation or precipitation protocols that preserve particle integrity and manage emulsifier residue.

2. METHODOLOGY

PRIMARY PATENT EVIDENCE

Patent 1

- Patent Number: EP2561011B1
- Patent Title: Thermoplastic and/or elastomeric composite material based on carbon nanotubes ...
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of agglomerated solid composite materials based on carbon nanotubes and graphenes in a thermoplastic and/or elastomeric polymer matrix. In the nitrile-related example, carbon nanotubes and graphenes are introduced into a BUSS co-kneader with a butadiene-acrylonitrile copolymer resin and an acrylic copolymer, followed by granulation and drying. - CNT content: 35 wt% — Example 7 - Graphene content: 5 wt% — Example 7 - Nitrile resin content: 55 wt% — Example 7 - Acrylic copolymer content: 5 wt% — Example 7 - Preheat temperature of butadiene-acrylonitrile copolymer: 160°C — Example 7 - Injection temperature of butadiene-acrylonitrile copolymer: 190°C — Example 7 - Co-kneader temperature: 200°C — Example 7 - Flow rate: 12 kg/h — Example 7 - Screw rotation speed: 240 rpm — Example 7 - Drying temperature: about 50°C — Example 7 - Co-kneader zone 1/zone 2 temperature: 280/290°C — Example 1 - Flow rate: 13 kg/h — Example 1 - Co-kneader zone 1/zone 2 temperature: 230-290°C — Example 2 - Twin-screw extruder diameter: 16 mm — Example 2 - Twin-screw extruder L/D: 25 — Example 2 - CNT content: 20 wt% — Example 3 - Graphene content: 5 wt% — Example 3 - PA-6 powder content: 75 wt% — Example 3 - PA-6 granules content: 5 wt% — Example 3 - Co-kneader zone 1/zone 2 temperature: 290/290°C — Example 3 - Flow rate: 11 kg/h — Example 3 - Co-kneader flow rate reduced to: 10 kg/h — Example 3 - Co-kneader zone 1/zone 2 temperature: 300/260°C and 310/270°C — Example 5 - CNT content: 15 wt% — Example 5 - Graphene content: 5 wt% — Example 5 - Polycarbonate powder content: 80 wt% — Example 5 - Polycarbonate granules content: 5 wt% — Example 5 - Flow rate: 10-11 kg/h — Example 5 - Flow rate: 15 kg/h — Example 5 - CNT content: 18 wt% — Example 11 - Graphene content: 2 wt% — Example 11 - Zone 1 temperature: 150°C — Example 11 - Zone 2 temperature: 140°C — Example 11 - Flow rate: 12 kg/h — Example 11 - Screw rotation speed: 200 rpm — Example 11

Relevant Experimental Evidence - Example 7: A nitrile-resin-based composite was produced by feeding CNTs, graphenes, and a butadiene-acrylonitrile copolymer into a BUSS co-kneader, yielding homogeneous granules. - Example 11: A solid nitrile elastomer resin base was compounded with CNTs and graphenes at 150/140°C and 12 kg/h to form granules containing 18 wt% CNT and 2 wt% graphene. - Example 3: Adding 5 wt% graphene to a PA-6/CNT system

reduced viscosity and stabilized continuous compounding relative to CNT-only formulation. - Example 5: Graphene addition enabled stable continuous processing at 15 kg/h without exceeding 300°C, unlike CNT-only formulation. - Example 6: Dilution of the composite masterbatch reduced resistivity by almost a decade in polycarbonate.

Technical Relevance This patent is only indirectly relevant to high-ACN NBR because it includes a butadiene-acrylonitrile copolymer as a matrix component in a conductive composite. It is useful as a processing reference for incorporating nitrile-based resins into compounded materials, but it does not disclose NBR polymerization or acrylonitrile-content control.

Patent 2

- Patent Number: EP2325238B1
- Patent Title: ... suspensions of functionalised, wholly or partially hydrated nitrile rubbers
- Assignee: Not disclosed
- Jurisdiction: EP
- Publication Year: 2017
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses preparation of aqueous suspensions of carboxylated, completely or partially hydrogenated nitrile rubber by precipitating HXNBR from acetone or THF solutions using water and amines such as diethylamine or dipropylamine. The process emphasizes low emulsifier content, controlled pH, and staged solvent removal to obtain stable suspensions with defined particle size. - HXNBR-1 solution concentration: 2.5 wt% — Example 9 - Water amount: 640 g — Example 9 - pH of water: 12.3 — Example 9 - Pressure: 250 mbar — Example 9 - Pressure range: 200-300 mbar — Example 9 - Wall temperature: 35°C — Example 9 - Wall temperature: 45°C — Example 9 - Wall temperature: 50°C — Example 9 - Pressure: 100 mbar — Example 9 - Total time: 7 hours 1 minute — Example 9 - Mean particle diameter: 229 nm — Example 9 - pH of latex: 9.8 — Example 9 - HXNBR-1 solution amount: 14.8 kg in 150 kg acetone — Example 10 - Water amount: 16 kg — Example 10 - Diethylamine amount: 150 g — Example 10 - Wall temperature: 25°C to 35°C — Example 10 - Wall temperature after 1 hour: 45°C — Example 10 - Pressure range: 300-400 mbar — Example 10 - Final solids content: 25.5% — Example 10 - Final pH: 10.0 — Example 10 - HXNBR-1 solution concentration: 3 wt% — Example 11 - Water amount: 640 g — Example 11 - Dipropylamine concentration: 0.2 mol/L — Example 11 - Temperature: 20°C — Example 11 - Particle diameter: 310 nm — Example 11 - Solids content: 27.1% — Example 11 - THF solution water ratio: 1:5 — Examples 12-26 - Diethylamine concentration: 0.2 mol/L — Examples 12-26 - Temperature: room temperature — Examples 12-26 - Pressure: normal pressure — Examples 12-26 - Solution composition: 10% HXNBR-1 in THF — Example 31 - Water amount: 120 g — Example 31 - Diethylamine concentration: 0.2 mol/L — Example 31 - Temperature: 40°C — Example 31 - Pressure: 150 mbar — Example 31 - Solids concentration: 37% — Example 31 - pH: 10.2 — Example 31 - Polymer solution flow: 5 kg/h — Example 32 - Water-amine mixture flow:

965 g/h — Example 32 - Wall temperature: 25°C — Example 32 - Wall temperature: 45°C — Example 32 - Pressure: 220 mbar abs — Example 32 - Diethylamine solution: 2.5 wt% — Example 32 - Diethylamine solution: 5 wt% — Example 32 - Polymer solution flow: 5 kg/h — Example 33 - Water flow: 1.2 kg/h — Example 33 - d50: 134 nm — Example 33 - d90: 281 nm — Example 33 - Wall temperature: 45°C — Example 33 - Pressure range: 230-270 mbar abs — Example 33 - Pressure range: 230-105 mbar abs — Example 33 - d50: 352 nm — Example 33 - d90: 732 nm — Example 33 - pH: 9.9 — Example 33 - Solution concentration: 10% HXNBR-2 or HXNBR-3 in THF — Examples 34-35 - Water amount: 120 g — Examples 34-35 - Diethylamine concentration: 0.2 mol/L — Examples 34-35 - Temperature: 40°C — Examples 34-35 - Pressure: 150 mbar — Examples 34-35 - Solids content: 30% — Example 18 - Centrifugation: 1200 g — Example 18 - Storage temperature: 40°C — Example 18 - Sample height: 52 mm — Example 18 - Solids content: 44.6% — Example 18 - pH: 9.2 — Example 18 - Particle diameter: 234 nm to 217 nm — Example 18

Relevant Experimental Evidence - Example 9: Precipitation from acetone with diethylamine-adjusted water produced a stable HXNBR latex with 229 nm mean particle size and pH 9.8. - Example 10: Repeated precipitation/evaporation cycles in acetone yielded a 25.5% solids HXNBR suspension at pH 10.0. - Example 11: THF precipitation with dipropylamine at 20°C gave a 310 nm suspension with 27.1% solids. - Example 31: THF precipitation at 40°C and 150 mbar produced a 37% solids suspension at pH 10.2. - Example 33: Continuous precipitation with deionized water and fed-batch workup produced suspensions with d50 134 nm and d90 281 nm, later 352/732 nm after further concentration.

Technical Relevance This patent is highly relevant as a downstream processing disclosure for hydrogenated nitrile rubber suspensions, showing how particle size, solids content, and pH can be controlled after polymer synthesis. While it does not teach high-acrylonitrile polymerization directly, it provides practical suspension/precipitation handling for nitrile-rubber-derived materials.

Patent 3

- Patent Number: US8299175B2
- Patent Title: Method of preparing thermoplastic resin having superior gloss, impact strength ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2012
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a two-step emulsion polymerization route for preparing an ABS graft copolymer latex and powder. A butadiene rubber latex is first prepared by staged emulsion polymerization with molecular weight control, then grafted with styrene and acrylonitrile, followed by coagulation at 60-80°C and aging at 80-99°C to obtain powder. - Rubber latex particle diameter: 1,800-5,000 Å — Claim 1 -

Polymerization conversion ratio: at least 90% — Claim 1 - Swelling index: 12-40 — Claim 1 - Average gel content: 70-95% — Claim 1 - Rubber latex amount: 45-75 parts by weight — Claim 1 - Styrene amount: 17-40 parts by weight — Claim 1 - Acrylonitrile amount: 4-20 parts by weight — Claim 1 - Primary reaction butadiene monomer: 50-95 parts by weight — Claim 1 - Primary reaction molecular weight controlling agent: 0.1-0.5 parts by weight — Claim 1 - Secondary reaction molecular weight controlling agent: 0.1-0.3 parts by weight — Claim 1 - Secondary reaction butadiene monomer: 5-50 parts by weight — Claim 1 - Primary reaction temperature: 60-75°C — Claim 1 - Primary reaction time: 8-25 hours — Claim 1 - Secondary reaction temperature: 70-85°C — Claim 1 - Secondary reaction time: 15-40 hours — Claim 1 - Coagulation temperature: 60-80°C — Claim 1 - Aging temperature: 80-99°C — Claim 1 - Example 1 butadiene monomer: 80 parts by weight — Example 1 - Example 1 potassium rosinate: 1.2 parts by weight — Example 1 - Example 1 potassium oleate: 1.5 parts by weight — Example 1 - Example 1 sodium carbonate: 0.7 parts by weight — Example 1 - Example 1 potassium bicarbonate: 0.8 parts by weight — Example 1 - Example 1 tert-dodecyl mercaptan: 0.3 parts by weight — Example 1 - Example 1 potassium persulfate: 0.3 parts by weight — Example 1 - Example 1 first reaction temperature: 68°C — Example 1 - Example 1 first reaction time: 13 hours — Example 1 - Example 1 remaining butadiene: 20 parts by weight — Example 1 - Example 1 second reaction temperature: 75°C — Example 1 - Example 1 second reaction time: 35 hours — Example 1 - Example 1 polymerization conversion ratio: 95% — Example 1 - Example 1 particle diameter: 2,700 Å — Example 1 - Example 1 gel content: 90% — Example 1 - Example 1 swelling index: 16 — Example 1 - Example 1 graft copolymer latex rubber latex: 60 parts by weight — Example 1 - Example 1 styrene: 19.2 parts by weight — Example 1 - Example 1 acrylonitrile: 8.2 parts by weight — Example 1 - Example 1 graft polymerization temperature: 70°C — Example 1 - Example 1 graft polymerization feed 2 styrene: 9.6 parts by weight — Example 1 - Example 1 graft polymerization feed 2 acrylonitrile: 3.0 parts by weight — Example 1 - Comparative Example 1 rubber latex: 76 parts by weight — Comparative Example 1 - Comparative Example 1 styrene: 8.2 parts by weight — Comparative Example 1 - Comparative Example 1 acrylonitrile: 4.2 parts by weight — Comparative Example 1 - Comparative Example 1 graft ratio: 24% — Comparative Example 1 - Comparative Example 2 rubber latex: 40 parts by weight — Comparative Example 2 - Comparative Example 2 styrene: 33.2 parts by weight — Comparative Example 2 - Comparative Example 2 acrylonitrile: 14.2 parts by weight — Comparative Example 2 - Comparative Example 2 graft ratio: 50% — Comparative Example 2 - Comparative Example 4 reaction temperature: 55°C — Comparative Example 4 - Comparative Example 4 reaction time: 10 hours — Comparative Example 4 - Comparative Example 4 reaction temperature: 65°C — Comparative Example 4 - Comparative Example 4 reaction time: 8 hours — Comparative Example 4 - Comparative Example 4 particle diameter: 1,000 Å — Comparative Example 4 - Comparative Example 4 gel content: 89% — Comparative Example 4 - Comparative Example 4 swelling index: 18 — Comparative Example 4 - Comparative Example 5 particle diameter: 5,100 Å — Comparative Example 5 - Comparative Example 5 graft ratio: 20% — Comparative Example 5 - Comparative Example 6 tert-dodecyl mercaptan primary: 0.6 parts by weight — Comparative Example 6 - Comparative Example 6 tert-dodecyl mercaptan secondary: 0.3 parts by weight — Comparative Example 6 - Comparative Example 6 particle diameter: 2,750 Å — Comparative Example 6 - Comparative Example 6 gel content: 60% — Comparative Example 6 - Comparative Example 7 rubber content: 13 wt% — Comparative Example 7 - Comparative Example 8 polymerization initiator:

water-soluble persulfate — Comparative Example 8 - Comparative Example 8 graft ratio: about 20% — Comparative Example 8 - Comparative Example 9 coagulation temperature: 55°C — Comparative Example 9 - Comparative Example 9 aging temperature: 75°C — Comparative Example 9 - Comparative Example 9 bulk density: 0.28 — Comparative Example 9 - Comparative Example 10 coagulation temperature: 85°C — Comparative Example 10 - Comparative Example 10 aging temperature: 92°C — Comparative Example 10

Relevant Experimental Evidence - Example 1: Two-step butadiene latex preparation achieved 95% conversion, 2,700 Å particle size, 90% gel content, and 16 swelling index. - Example 1: Subsequent grafting with styrene and acrylonitrile produced ABS graft latex for final resin preparation. - Comparative Example 4: Small-particle latex (1,000 Å) had high gel but limited impact strength. - Comparative Example 5: Very large particles (5,100 Å) lowered graft ratio and gloss. - Comparative Example 8: Using water-soluble persulfate led to yellowing on drying.

Technical Relevance This patent is relevant because it explicitly controls acrylonitrile feed in an emulsion graft system and shows how monomer staging, particle size, and gel content affect final resin properties. Although it is an ABS patent rather than NBR, the staged emulsion polymerization principles are directly informative for high-acrylonitrile nitrile-rubber synthesis and composition control.

Patent 4

- Patent Number: US9388302B2
- Patent Title: Latex for dip molding, composition for dip molding, preparation method of dip ...
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2016
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses a latex for dip molding based on a conjugated diene monomer, an ethylenically unsaturated nitrile monomer, and an ethylenically unsaturated acid monomer, optionally with sodium methallyl sulfonate as a copolymerizable ionic monomer. The latex is used in a dip-molding process involving coagulant deposition, dipping, drying, leaching, and heat curing to form antistatic and mechanically robust glove products. - Sodium methallyl sulfonate: 2 parts by weight — Example 1 - Acrylonitrile: 30 wt% — Example 1 - 1,3-butadiene: 66 wt% — Example 1 - Methacrylic acid: 4 wt% — Example 1 - Sodium alkyl benzene sulfonate: 2 parts by weight — Example 1 - t-dodecyl mercaptan: 0.5 parts by weight — Example 1 - Deionized water: 140 parts by weight — Example 1 - Potassium persulfate: 0.3 parts by weight — Example 1 - Sodium dimethyldithiocarbamate: 0.1 parts by weight — Example 1 - Polymerization temperature: 40°C — Example 1 - Conversion rate: 95% — Example 1 - Solid content: 45% — Example 1 - pH: 8.5 — Example 1 - Composition solid content: 25% — Example 1 - Composition pH: 10.0 — Example 1 - Sulfur: 1.5 parts by weight — Example

1 - Zinc oxide: 1.5 parts by weight — Example 1 - Zinc dibutyldithiocarbamate: 0.5 parts by weight — Example 1 - Coagulant calcium nitrate: 22 parts by weight — Example 1 - Coagulant methanol: 69.5 parts by weight — Example 1 - Coagulant calcium carbonate: 8 parts by weight — Example 1 - Coagulant wetting agent: 0.5 parts by weight — Example 1 - Drying temperature: 70°C — Example 1 - Cross-linking temperature: 125°C — Example 1 - Cross-linking time: 20 minutes — Example 1 - Sodium methallyl sulfonate: 5 parts by weight — Example 2 - Sodium methallyl sulfonate: 10 parts by weight — Example 3 - Sodium alkyl benzene sulfonate: 1 part by weight — Example 4 - Acrylonitrile: 35 wt% — Example 5 - 1,4-butadiene: 61 wt% — Example 5

Relevant Experimental Evidence - Example 1: A 30 wt% acrylonitrile latex with sodium methallyl sulfonate was prepared and converted into a glove-dip composition with 25% solids and pH 10.0. - Example 1: Coagulant dipping at 70°C drying and 125°C crosslinking produced a glove-shaped dip molded product. - Example 2: Increasing sodium methallyl sulfonate to 5 parts by weight was evaluated against Example 1. - Example 3: Increasing sodium methallyl sulfonate to 10 parts by weight was evaluated against Example 1. - Example 5: Raising acrylonitrile to 35 wt% and lowering butadiene to 61 wt% provided an alternative high-ACN latex formulation.

Technical Relevance This patent is highly relevant because it discloses nitrile-butadiene latex compositions with explicit acrylonitrile levels of 30 and 35 wt%, which fall in the high-ACN range. It also shows how ionic monomer addition, emulsifier choice, and dip-molding/coagulation conditions influence latex stability and final glove performance.

Patent 5

- Patent Number: US20200299532A1
- Patent Title: Antistatic gloves and process for making same
- Assignee: Not disclosed
- Jurisdiction: US
- Publication Year: 2020
- Legal Status: Unknown
- Polymer Type: Not disclosed
- Relevance: Automatically extracted candidate

Polymerization / Synthesis Method - Synthesis method: This patent discloses antistatic elastomeric articles, including multilayer gloves made from blends of nitrile rubber latex and natural rubber latex. The process prepares separate latex blends for outer and inner layers, then forms the glove by double-dip coagulant dipping, chlorination, washing, drying, and curing. - Surface resistivity threshold: below about 10^{14} Ω /sq — Abstract - Static decay time threshold: less than about 60 seconds — Abstract - Outer layer nitrile rubber: 80% — Example 2 - Outer layer natural rubber: 20% — Example 2 - Inner layer natural rubber: 80% — Example 2 - Inner layer nitrile rubber: 20% — Example 2 - Former temperature: 59-66°C — Example 2 - Coagulant temperature: 53-59°C — Example 2 - Drying temperature: 70°C — Example 2 - First latex dwell time: 12-18 seconds — Example 2 - Second latex dwell time: 5-10 seconds — Example 2 - Intermediate drying temperature: 135°C — Example 2 - Intermediate drying time: 1 minute —

Example 2 - Leaching temperature: 40°C — Example 2 - Leaching time: 3 minutes — Example 2 - Final curing temperature: 135°C — Example 2 - Final curing time: 20 minutes — Example 2 - Chlorine solution concentration: about 300 ppm — Example 2 - Chlorination time: 35 minutes — Example 2 - Outer layer nitrile rubber: 95% — Example 3 - Outer layer natural rubber: 5% — Example 3 - Inner layer natural rubber: 95% — Example 3 - Inner layer nitrile rubber: 5% — Example 3 - Surface resistivity: $3.3 \times 10^{12} \Omega/\text{sq}$ — Example 3 - Static decay time: 22 seconds — Example 3 - Modulus at 500% elongation: 3.6 MPa — Example 3 - Ultimate elongation: 609% — Example 3 - Outer layer nitrile rubber: 80% — Example 4 - Outer layer natural rubber: 20% — Example 4 - Inner layer natural rubber: 80% — Example 4 - Inner layer nitrile rubber: 20% — Example 4 - Surface resistivity: $1.7 \times 10^{13} \Omega/\text{sq}$ — Example 4 - Static decay time: 27.2 seconds — Example 4 - Modulus at 500% elongation: 3.7 MPa — Example 4 - Ultimate elongation: 569% — Example 4 - Outer layer nitrile rubber: 60% — Example 5 - Outer layer natural rubber: 40% — Example 5 - Inner layer natural rubber: 60% — Example 5 - Inner layer nitrile rubber: 40% — Example 5 - Surface resistivity: $2.8 \times 10^{13} \Omega/\text{sq}$ — Example 5 - Static decay time: 29.2 seconds — Example 5 - Modulus at 500% elongation: 4.2 MPa — Example 5 - Ultimate elongation: 569% — Example 5 - Outer layer nitrile rubber: 51% — Example 6 - Outer layer natural rubber: 49% — Example 6 - Inner layer natural rubber: 51% — Example 6 - Inner layer nitrile rubber: 49% — Example 6 - Surface resistivity: $3.0 \times 10^{13} \Omega/\text{sq}$ — Example 6 - Static decay time: 34.0 seconds — Example 6 - Modulus at 500% elongation: 6.6 MPa — Example 6 - Ultimate elongation: 569% — Example 6 - Coagulant calcium nitrate: 12.00 wt% — Table 5 - Coagulant calcium carbonate: 4.80 wt% — Table 5 - Coagulant water: 82.95 wt% — Table 5 - Cellosize QP52,000: 0.05 wt% — Table 5 - Surfynol Tg: 0.20 wt% — Table 5 - Specific gravity of coagulant: 1.080-1.090 — Table 5

Relevant Experimental Evidence - Example 2: A double-dip process produced a two-layer glove with 80/20 nitrile/natural rubber layering and antistatic performance. - Example 3: A 95/5 nitrile/natural rubber laminate achieved $3.3 \times 10^{12} \Omega/\text{sq}$ surface resistivity and 22 s static decay time. - Example 4: An 80/20 laminate remained delamination-free after chlorination and retained antistatic properties. - Example 5: A 60/40 laminate maintained antistatic behavior with higher modulus than Example 4. - Example 6: A near-balanced 51/49 laminate still met antistatic targets but with higher modulus.

Technical Relevance This patent is relevant because it uses nitrile rubber latex in multilayer glove structures and demonstrates how nitrile content affects antistatic performance and mechanical feel. While not a synthesis patent for NBR itself, it provides composition and processing guidance for high-nitrile latex blends used in end-use articles.

3. CROSS-PATENT COMPARISON & SYNTHESIS TRENDS

- EP2325238B1 focuses on post-synthesis precipitation and suspension stabilization of HXNBR, whereas US9388302B2 and US20200299532A1 focus on latex composition and article formation rather than polymer backbone synthesis.

- US9388302B2 is the clearest high-acrylonitrile disclosure, with 30 wt% and 35 wt% acrylonitrile in nitrile-butadiene latex formulations; US20200299532A1 uses nitrile rubber blends but does not disclose acrylonitrile content.
- US8299175B2 uses staged emulsion polymerization with 4-20 parts by weight acrylonitrile in an ABS graft system, showing monomer-feed control principles but not an NBR product.
- Emulsion polymerization is the dominant synthesis route across the relevant patents; EP2325238B1 instead emphasizes solvent precipitation of already-formed HXNBR into aqueous suspension.
- Emulsifier and chain-transfer control are explicit in US8299175B2 and US9388302B2; EP2325238B1 emphasizes amine-assisted precipitation and pH control; US20200299532A1 emphasizes latex blending, coagulant dipping, chlorination, and curing.
- The most relevant high-ACN composition window in the evidence is 30-35 wt% acrylonitrile in US9388302B2, which aligns with high-ACN NBR definitions and is more directly applicable than the ABS or composite patents.

4. CONCLUSION

The evidence shows that high-acrylonitrile NBR development is best supported by emulsion-polymerized latex systems with explicit monomer-ratio control, staged feed strategies, and careful downstream coagulation or dip-molding processing. Among the patents reviewed, US9388302B2 provides the most direct high-ACN NBR composition evidence, with 30 wt% and 35 wt% acrylonitrile formulations, while US8299175B2 reinforces the importance of staged monomer addition, particle size, gel content, and chain-transfer control in nitrile-containing emulsion systems. EP2325238B1 contributes practical precipitation/suspension handling for HXNBR-derived materials, and US20200299532A1 shows how nitrile-rich latex blends can be converted into antistatic articles with acceptable mechanical properties. Recommended evidence-based synthesis parameters are: emulsion polymerization; acrylonitrile in the 30-35 wt% range for high-ACN grades; butadiene as the major comonomer; controlled use of emulsifier and mercaptan chain-transfer agent; staged monomer addition to manage particle size and gel; and coagulation/drying conditions matched to the intended latex or article format.

5. REFERENCES

- EP2561011B1 | Thermoplastic and/or elastomeric composite material based on carbon nanotubes ... | Not disclosed | EP | 2016 | <https://patents.google.com/patent/EP2561011B1/en>
- EP2325238B1 | ... suspensions of functionalised, wholly or partially hydrated nitrile rubbers | Not disclosed | EP | 2017 | <https://patents.google.com/patent/EP2325238B1/en>
- US8299175B2 | Method of preparing thermoplastic resin having superior gloss, impact strength ... | Not disclosed | US | 2012 | <https://patents.google.com/patent/US8299175B2/en>
- US9388302B2 | Latex for dip molding, composition for dip molding, preparation method of dip ... | Not disclosed | US | 2016 | <https://patents.google.com/patent/US9388302B2/en>

- US20200299532A1 | Antistatic gloves and process for making same | Not disclosed | US | 2020 | <https://patents.google.com/patent/US20200299532A1/en>